Yudi (Mike) Qin

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Education

University of Illinois at Urbana-Champaign Aug. 2022 - Dec. 2024
M.S. in Computer Science | Thesis Advisor: Nancy M. Amato

University of Illinois at Urbana-Champaign Aug. 2018 - May 2022
B.S. in Computer Science + Geography & GIS | GPA: 3.95/4.0

Research Interests

Robotics, Task and Motion Planning, Multi-Robot Systems, Learning for Planning

Publications

Peer Reviewed Journal

- **M. Qin**, I. Solis, J. D. Motes, M. Morales and N. M. Amato, "K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning," in IEEE Robotics and Automation Letters, vol. 10, no. 10, pp. 10562-10569, Oct. 2025.
- I solis, J, Motes, **M. Qin**, M. Morales and N. M. Amato,"Experience-based Subproblem Planning for Multi-Robot Motion Planning". 2025. In Submission.
- I Solis, J. Motes, **M. Qin**, M. Morales and N. M. Amato,"Adaptive Robot Coordination: A Subproblem-Based Approach for Hybrid Multi-Robot Motion Planning", in IEEE Robotics and Automation Letters, vol. 9, no. 8, pp. 7238-7245, Aug. 2024.

Thesis and Others

- **M. Qin**, I. Solis, J. D. Motes, M. Morales and N. M. Amato, "A Hyrbid Framework for Multi-Robot Kinodynamic Planning," Master's Thesis. Champaign, IL. Dec. 2024.

Experience

Ubtech Robotics Inc. Feb. 2025 - Present
Robotics Algorithm Engineer, Autonomous Navigation Beijing, China

- Designed and implemented a free-space planning module in addition to the existing autonomous navigation stack while regularly maintaining the codebase.
- Implemented planning algorithms from state-of-the-art research papers in the freespace-planning module, focusing on improving the overall accuracy and success rate in autonomous parking scenarios.
- Developed an Isaac Sim simulation framework alongside the existing Webots simulation framework, which supports the simulation of modules including perception, localization and mapping, planning and control.

Parasol Lab Aug. 2022 - Present
Graduate Research Assistant Champaign, IL

- Contributed to the development of the multi-robot system components for PPL(Parasol Planning Library), an open-source C++ robotics task and motion planning library
- Developed an algorithm for solving multi-robot motion planning problems, achieving an overall 20% speed-up over existing solutions across various test scenarios.
- Extending the algorithm to incorporate trajectory optimzations for real-time controls of the robots.

Foxconn Interconnect Technology May 2023 - Aug. 2023
Software Development Engineering Intern Remote

- Developed a software package in ROS for motion planning for multiple AGV(Automated Guided Vehicle) robots.
- Worked closely with my mentor to put this package into real usage for the company's factory automation.
- Recognized as "Excellent Intern" for consistent contributions and performance during the internship.

Parasol Lab May 2021 - Aug. 2021
Research Experience for Undergraduate (REU) Student Champaign, IL

- Assisted in developing and implementing multi-robot motion planning algorithms in **C++**.
- Contributed to developing and open-sourcing PPL. Specifically, added testing suites for existing codes, refactored existing codes, and added new components to the library.

CyberGIS Center Jun. 2020 - Mar. 2022
Student Intern Champaign, IL

- Developed Python codes for processing large-scale geo-located Twitter data, gathered using Twitter's data collection API.
- Contributed to developing a web-based GIS application that is able to visualize and predict spatiotemporal patterns of infectious diseases.
- Implemented a spatial analytic tool known as MGWR (multi-scale geographically weighted regression) in Python and explored its application on disease data analysis.

Teaching Experience

University of Illinois at Urbana-Champaign Aug. 2022 - Present
Graduate Teaching Assistant Champaign, IL

- CS 441, Applied Machine Learning (Fall 2022)
- CS 421, Programming Languages and Compilers (Spring and Summer 2023)
- CS 440, Intro to Artificial Intelligence (Spring 2024)

Honors and Awards

UIUC College of Liberal Arts and Science, Dean's List Fall 2018 - Spring 2022
The Howard and Ruth Roepke Undergraduate Scholarship Spring 2019